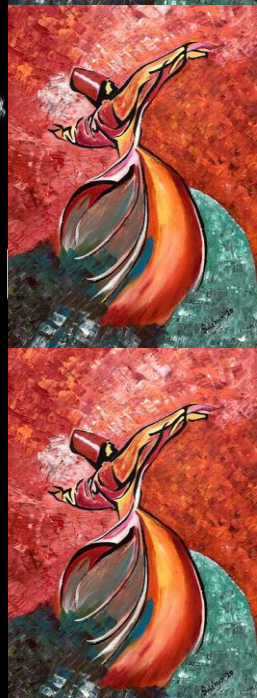




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



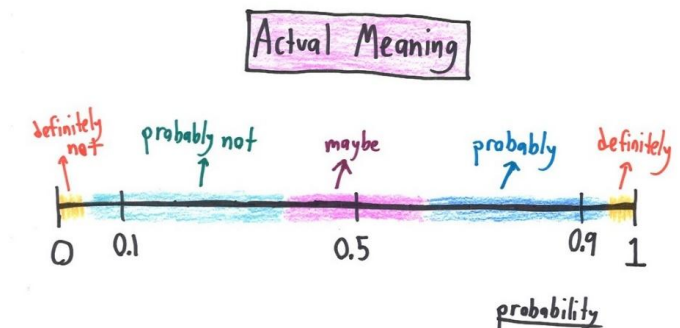


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 6 - Lecture No 10 – 22nd September 2025



Announcements!

1. Lectures up to No 09 posted
2. An added lecture will be posted in a asynchronous mode in lieu of September 08, 2025 class
3. Quiz No 03 – September 24, 2025 (Wednesday, Lecture No 1) on Counting
4. TA's

Agenda for today

1. Unit 4 - Discrete Random Variables



Poll for MT
1 questions

[Change Kahoot](#)

ⓘ Change to Slide Show View to play this kahoot.



Meet Your TAs Saif Nazir

About Me

I am a Computer Engineering Senior. I am interested in Robotics and AI. Apart from Academics, I play Tabla and Sitar and I am also part of the volleyball team at Habib. You can always find me either in the Music room or at the courts.

Ehsaas Hours

- * Tue 12:00 – 1:30
- * Thu 12:00 – 1:30

Contact Me

✉ sno8448@st.habib.edu.pk
or “Saif Nazir” on Teams



Meet Your TAs Rayyan Shah

About Me

I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours

- * Mon 11:30 – 12:30
- * Tue 2:30 – 3:30
- * Fri 3:30 – 4:30

Contact Me

✉ rso8770@st.habib.edu.pk
Or “Rayyan Ali Shah” on Teams



Meet Your TAs Hiba Aiman Ali

About Me

I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I'm not in class, I'm most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours

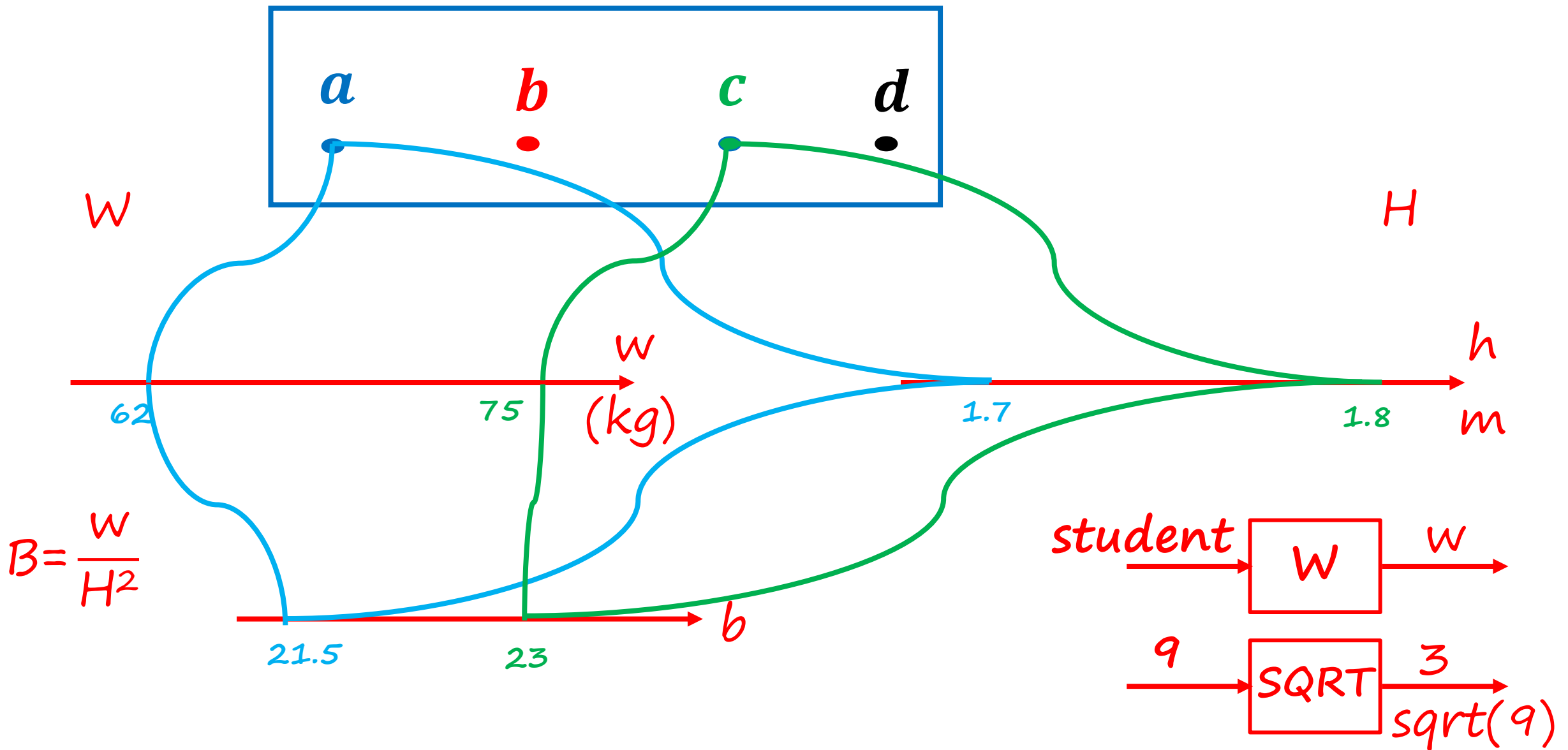
- * Tue 4:00 – 5:00
- * Fri 8:30 – 10:30

Contact Me

✉ ha09269@st.habib.edu.pk
on Teams or Outlook

Unit 4
Discrete Random Variables

Random Variables – The Idea



Random Variables – The formalism:

- A random variable (“r .v”) associates a value (a real number) to every possible outcome.
- **Mathematically:** A function from the sample space Ω to the real numbers.
- It can take discrete or continuous values

Notation: random variable X numerical value x

- We can have several random variables defined on the same sample space
- A function of one or several random variables is also a random variable

❖ Meaning of $X + Y$:

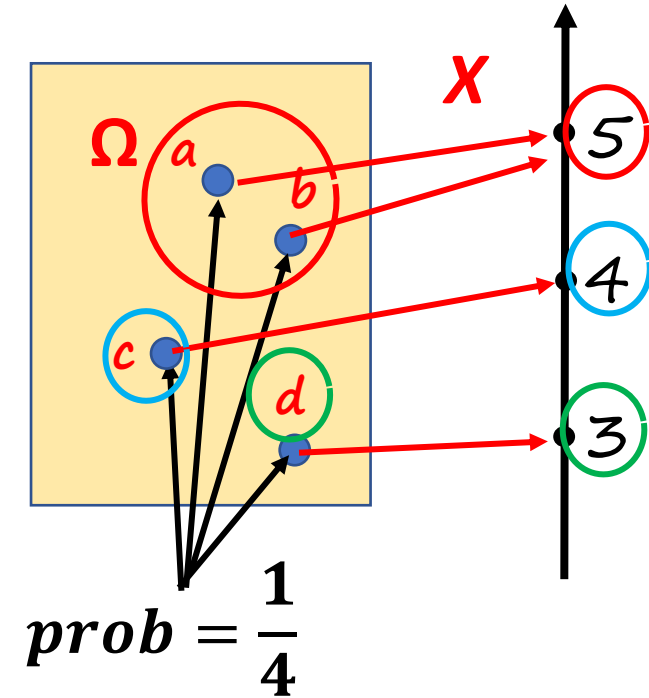
*r.v takes value $x+y$.
When X takes value x and Y takes value y*

Probability mass function (PMF) of a discrete r.v. X

- It is the “probability law” or “probability distribution” of X
- If we fix some x , then “ $X = x$ ” is an event

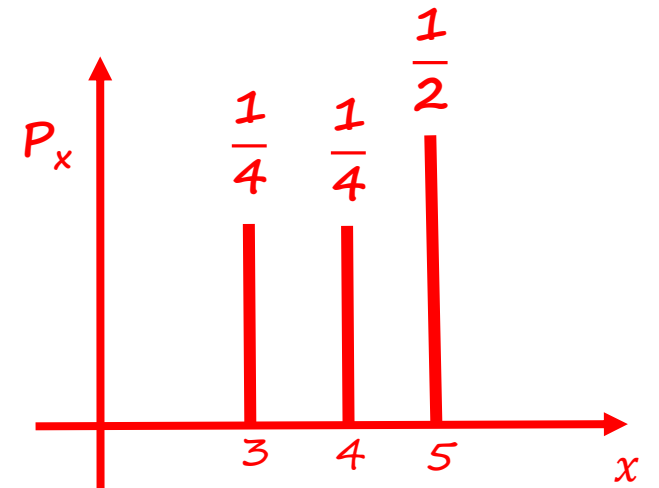
$$\rightarrow x=5, \quad X=5 \quad \{w : X(w) = 5\} = \{a, b\}$$

$$\rightarrow P_x(5) = \frac{1}{2}$$



$$p_X(x) = \mathbf{P}(X = x) = \mathbf{P}(\{w \in \Omega \text{ s.t. } X(w) = x\})$$

$$\bullet \text{ Properties: } p_X(x) \geq 0 \quad \sum_x p_X(x) = 1$$



Sample space: discrete/finite example

- Two rolls of a tetrahedral die

Y =
Second
roll

4	5	6	7	8
3	4	5	6	7
2	3	4	5	6
1	2	3	4	5
	1	2	3	4

X = First roll

- Let every possible outcome have probability $1/16$

$$\diamond Z = X + Y$$

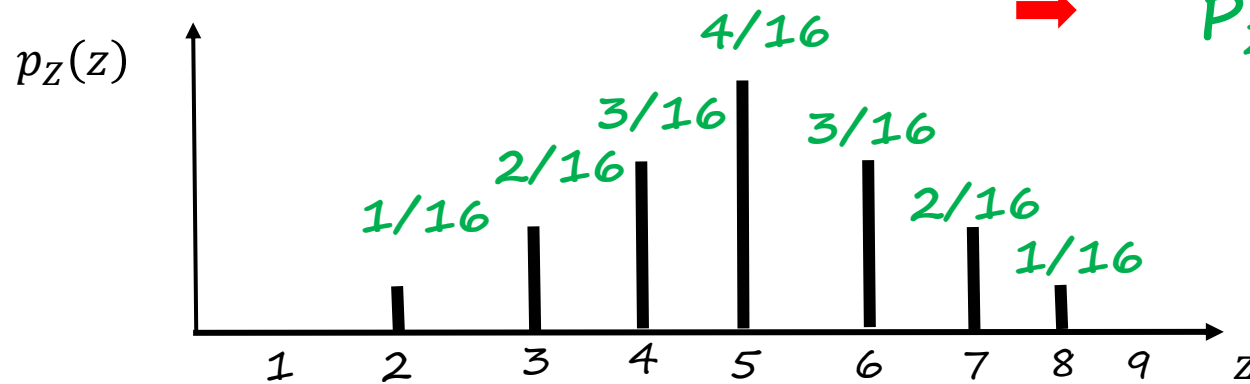
Find $p_Z(z)$ for all z

- repeat for all z :
 - ✓ collect all possible outcomes for which Z is equal to z
 - ✓ add their probabilities

$$\rightarrow P_Z(2) = P(Z = 2) = 1/16$$

$$\rightarrow P_Z(3) = P(Z = 3) = 2/16$$

$$\rightarrow P_Z(4) = P(Z = 4) = 3/16$$



a short review!

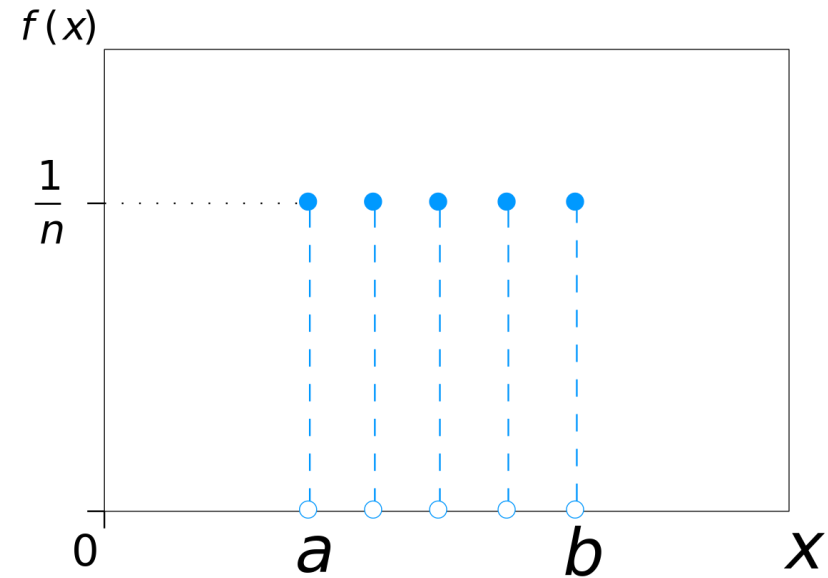
Bernoulli Random Variable



parameters

p

Discrete Uniform Random Variable



$$1/n = 1/(b - a + 1)$$

parameters

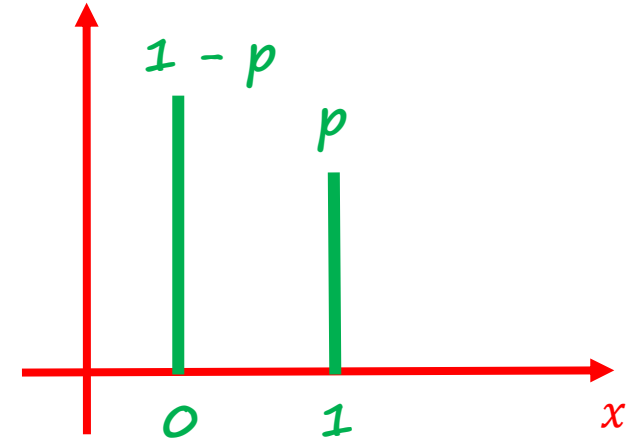
a, b

The simplest Random variable; Bernoulli – parameters $p \in [0, 1]$

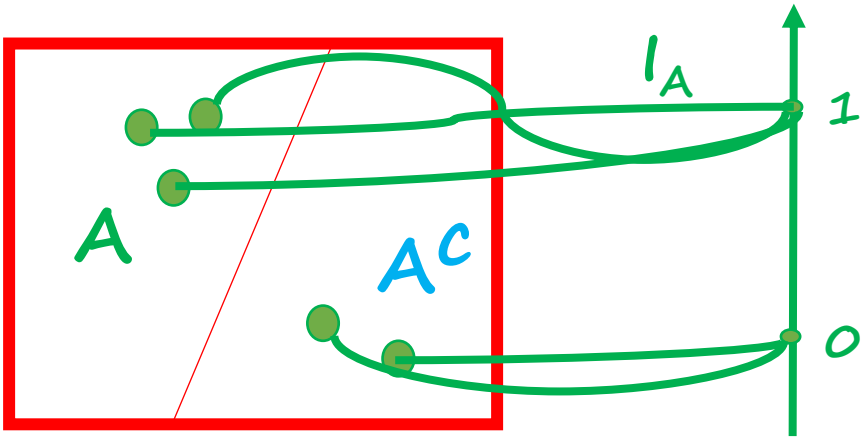
$$X = \begin{cases} 1, & \text{w.p. } p \\ 0, & \text{w.p. } 1 - p \end{cases}$$

$$P_X(0) = 1 - p$$

$$P_X(1) = p$$



- Models a trial that results in success/failure, Heads/Tails, etc.
- Indicator r.v. of a event A : $I_A = 1$ iff A occurs



$$P_{I_A}(1) = P(I_A = 1) = P(A)$$

$\uparrow$
 p

Discrete Uniform Random variable – parameters a, b

- **Parameters:** integers a, b
- **Experiment:** Pick one of $a, a+1, \dots, b$ at random; all equally likely
- **Sample Space:** $\{a, a+1, \dots, b\}$ $b - a + 1$ possible values
- **Random Variable X :** $X(\omega) = \omega$ $11:52:29$ $\{0, 1, \dots, 59\}$
- **Model of:** complete ignorance

Special case: $a = b$

